

BPG-10 · BEST PRACTICE GUIDES

Contingency and management reserve

Two reserves, two owners, two entirely different governance events

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— Summary

Contingency and management reserve are not two sizes of the same pot. This guide sets out what each covers, who owns it, where it sits relative to the cost baseline, how contingency is derived from and kept traceable to the risk register, how a drawdown is approved and recorded without moving the budget at completion, and how to read the drawdown curve as an early-warning indicator months before the forecast catches up.

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Contingency and management reserve

In one paragraph. Contingency and management reserve are not two sizes of the same pot. This guide sets out what each covers, who owns it, where it sits relative to the cost baseline, how contingency is derived from and kept traceable to the risk register, how a drawdown is approved and recorded without moving the budget at completion, and how to read the drawdown curve as an early-warning indicator months before the forecast catches up.

Who this is for. Project controls managers, cost engineers, risk managers and project sponsors — and anyone who has been asked "how much contingency is left" and found that the honest answer took a week to assemble.

— 1. Two reserves answer two different questions

The distinction is not about size or seniority. It is about what kind of uncertainty each reserve is funding.

Contingency reserve funds **identified** risk — the entries in the risk register, plus quantified estimating uncertainty. Someone has written the risk down, assessed it and priced it. Contingency sits **inside** the cost baseline, is controlled by the project, and is drawn down as those identified risks materialise.

Management reserve funds **unidentified** risk — the events nobody wrote down, and scope not yet foreseen. It sits **outside** the cost baseline, is controlled by the sponsor or the governing body, and its release is a change to the baseline rather than a movement within it.

The budget structure follows directly:

Cost baseline (BAC) = Σ control account budgets + contingency reserve
Total authorised budget = BAC + management reserve

Everything that matters operationally falls out of that arithmetic. Earned value performance is measured against BAC, so contingency is inside the measurement and management reserve is not. Drawing contingency does not change BAC. Releasing management reserve does.

Note also what the two reserves are not. Neither is an accounting provision. How reserves, provisions and onerous-contract positions are recognised in financial statements is governed by the applicable reporting framework and varies; a project budget reserve is a control instrument, and the accounting treatment is a separate question for the finance function.

— 2. Where contingency comes from

Contingency has one legitimate parent: **analysed risk**. A percentage applied to the estimate is not contingency, it is a habit. The moment anyone asks "what is this covering", a percentage has no answer, and the reserve becomes indefensible in exactly the meeting where it needs defending.

Two derivations are standard, and they answer different questions.

Expected monetary value (EMV) sums *probability × impact* across the register. It gives the mean outcome — the amount you would need if this project were run many times. It is simple, transparent and traceable to individual risks, and it systematically underfunds any single project, because a single project does not experience the mean.

A probabilistic P-level, typically from a Monte Carlo simulation of the register, gives the amount at which the project has a stated confidence of not overrunning — commonly P80, meaning an 80 % chance the outturn falls at or below the funded level. It captures how risks combine, including the effect of shared drivers, which the EMV sum cannot.

The gap between the two is the price of confidence, and stating it is good practice: "the mean exposure is 1.74 million; funding to P80 costs 2.60 million; the 0.86 million difference buys the organisation's stated risk appetite." That sentence turns a reserve from a number into a decision the sponsor has actually taken.

Correlation matters here more than most registers admit. Risks driven by a single shared cause — one supplier, one permitting authority, one commodity — do not behave independently, and treating them as independent understates the tail without changing the mean. Quantification method belongs to *BPG-17 – Quantitative schedule risk analysis* and the register itself to *BPG-16 – Risk registers that work*; what this guide insists on is the link: **every unit of contingency should be traceable to a risk identifier**.

— 3. Traceability, and why it is the whole control

A contingency line that cannot be decomposed into risks is a slush fund, and it will be spent like one. Traceability delivers four things that no aggregate figure can:

1. **A drawdown test.** A request to draw contingency is answerable — does it correspond to a risk in the register, and did that risk occur? If not, the cost is an overrun, and calling it a contingency draw conceals a performance problem inside a risk provision.
2. **A release mechanism.** When a risk passes without occurring, its contingency can be released back to the pot or to the sponsor, deliberately and visibly.
3. **A re-assessment path.** When a risk's probability or impact changes, the required contingency changes with it, and the reserve is re-derived rather than argued about.
4. **A defence.** "This is what it covers, risk by risk" is the only answer that survives an assurance review.

The practical form is unglamorous: a contingency register keyed to risk identifiers, showing for each risk the assessed exposure, the amount allocated, the amount drawn, the residual, and the status. It is one extra column set on the risk register rather than a separate document, and it should reconcile to the total contingency figure in the budget every month.

— 4. Holding it, and drawing it down

Hold contingency centrally. Contingency distributed into control account budgets stops being contingency; it becomes budget, and budget is always eventually consumed. The control account manager who has 60,000 of "risk allowance" inside a 900,000 package has no incentive to hand it back and no mechanism to do so. Hold the reserve at project level and release it against evidence.

Define the drawdown mechanism before the first draw. It needs four elements:

- **A trigger** — the risk has occurred, or has become certain to occur.
- **Evidence** — what happened, which risk identifier it corresponds to, and the cost of the consequence.
- **An approval level** — thresholds by value, with the sponsor's involvement rising with the amount.
- **A record** — a numbered drawdown entry, posted through the change log so the movement is auditable.

Record the draw as a transfer, not a variance. This is the mechanical point most often muddled. When contingency is drawn, budget moves within the baseline: the contingency pot decreases and a control account budget increases by the same amount. BAC is unchanged. Planned value for the affected account increases from the point of transfer forward, so the account's future earned value measurement is against the revised budget. Nothing about the past is restated. If your process instead increases BAC when contingency is drawn, contingency was never inside the baseline, and every historical index is irreconcilable.

— 5. The drawdown curve as an early-warning indicator

Plot cumulative contingency drawn against time, with two comparators on the same axes: physical progress (percent complete by earned value) and, separately, the remaining assessed exposure. The shape tells you things the cost report will not say for another two quarters.

Healthy. Drawdown tracks the retirement of risk. Contingency falls as risks either occur (drawn) or pass (released), and the remaining reserve stays at or above the remaining exposure throughout.

Front-loaded. Contingency consumed rapidly in the first third. Either the early risks were the severe ones — legitimate, and the register should show it — or contingency is subsidising ordinary estimating error, which will continue. A front-loaded curve on a project whose register still carries its major risks is one of the strongest early-warning signals in project controls, because it means the reserve will be gone before the exposure is.

Flat then cliff. Nothing drawn for months, then a large single draw. Almost always a reporting failure rather than a risk profile: costs were absorbed into control accounts and only escalated when they became impossible to absorb. Look for the corresponding CPI deterioration in the preceding periods.

Never drawn. Either the risks are not materialising — say so, and consider releasing reserve — or draws are being avoided because the approval process is painful, in which case the same costs are hiding inside control accounts and the contingency figure is meaningless.

The test that matters is not the burn ratio. Comparing "percentage of contingency drawn" with "percentage complete" is the intuitive check and it is close to useless, because contingency is not consumed in proportion to progress; it is consumed in proportion to risk occurrence, which is lumpy. The defensible test is:

Remaining contingency $\geq$ Remaining exposure (at the same confidence level as it was set)

Run it every month. §8 shows a project that passes the burn-ratio test comfortably and fails this one.

— 6. Management reserve: release is a baseline change

Management reserve is used when something occurs that the register did not and reasonably could not contain — a new statutory requirement, a client-directed change of a kind nobody anticipated, a systemic event. Three consequences follow, and all three are frequently missed.

It is a re-baselining event. Releasing reserve into the baseline increases BAC. Every earned value measure computed against the old BAC is now on a different basis, and the change must be logged with its date, so a reader of the trend knows where the discontinuity is. [BPG-04 – Baselining and baseline change control](#) owns the procedure.

It is a governance decision, not a controls decision. The project may request; the sponsor or governing body releases. A project that can draw management reserve on its own authority does not have a management reserve.

It is not a solution to an exhausted contingency. When the register has outgrown the reserve, the correct response is to re-derive contingency from the current register, show the shortfall, and ask for a decision — funding it, de-scoping, or accepting the exposure. Silently drawing management reserve to cover identified risks converts a visible funding gap into an invisible one, and destroys the distinction that makes either reserve meaningful.

— 7. Schedule contingency, briefly

The same logic applies in time. Schedule contingency is the float deliberately held against identified schedule risk, owned by the project and drawn down as risks occur; it is not the incidental float that falls out of the network, and it should be visible as a named buffer rather than hidden in inflated durations. Padding individual activities is the schedule equivalent of distributing contingency into control accounts: the protection is consumed silently and cannot be managed. Quantification of schedule risk is the subject of [BPG-17 – Quantitative schedule risk analysis](#).

Where the two interact — a delay that draws both time and time-related cost — take care not to fund the same event twice, once as schedule contingency and again as cost contingency for prolongation. Price it once, in the register, against one risk identifier.

— 8. How this goes wrong

Contingency set as a percentage. Ten per cent, because it has always been ten per cent. It is undefendable, it bears no relation to this project's risk profile, and it is the first thing an assurance reviewer will ask about.

Contingency distributed into control accounts. It is spent. Always.

Drawdown without a risk identifier. Ordinary overruns re-labelled as risk events. The reserve depletes, the CPI looks better than it should, and the register never changes — a combination that should trigger immediate investigation.

Double-counting risk and escalation. A commodity price risk priced in the register and covered by an escalation allowance in the estimate. Both are legitimate instruments; funding the same exposure twice is not, and it inflates the reserve in a way that is easy to attack.

Treating unspent contingency as a saving. At completion, contingency not drawn should be explained, not celebrated. Either the register overstated the exposure, which is an estimating lesson worth capturing, or the risks genuinely did not materialise, which is luck worth distinguishing from performance.

Reporting the pot without the exposure. "Contingency remaining: 1.65 million" is not information. "1.65 million remaining against 1.90 million of assessed remaining exposure" is.

Using contingency to fund scope. Scope growth is a change, and it belongs in the change control process with a decision about who pays. Absorbing it in contingency hides the growth and defunds the risks the reserve exists for.

No release discipline. Risks pass, and their contingency stays in the pot without being re-assessed. Over time the reserve stops corresponding to the register in either direction, and nobody trusts the figure.

— 9. Worked example

Illustrative figures. Currency USD; a 20-month project; risk impacts are the assessed cost consequence should the risk occur; the confidence level for contingency is P80, taken from a Monte Carlo model of the register; all figures at the stated data dates.

At sanction — assembling the budget.

Element	Amount
Σ control account budgets	24,000,000
Contingency reserve (set at P80)	2,600,000
Budget at completion (BAC)	26,600,000
Management reserve (sponsor-held, outside baseline)	1,400,000
Total authorised budget	28,000,000

Deriving the contingency. The register carries five priced risks:

Risk	Description	Probability	Impact	EMV = P × I
R-01	Consent granted later than programmed	0.40	900,000	$0.40 \times 900,000 = 360,000$
R-02	Ground conditions worse than survey	0.30	1,500,000	$0.30 \times 1,500,000 = 450,000$
R-03	Steel price movement beyond the hedged volume	0.50	700,000	$0.50 \times 700,000 = 350,000$
R-04	Specialist subcontractor default	0.15	2,000,000	$0.15 \times 2,000,000 = 300,000$
R-05	Rework arising from late design change	0.35	800,000	$0.35 \times 800,000 = 280,000$
Total EMV (the mean)				1,740,000

The Monte Carlo model of the same register, capturing the shared exposure between R-02 and R-05 (both driven by the same late design package), returns a P80 of **2,600,000**. Contingency is set there.

$$\text{Cost of confidence} = \text{P80} - \text{EMV} = 2,600,000 - 1,740,000 = 860,000$$

That 860,000 is the amount the organisation is paying to move from funding the average outcome to funding an 80 % confidence outcome. It is a sponsor's decision, and it is recorded as one.

At Month 9 — the drawdown position.

Two risks have occurred. R-01 materialised and the consent delay cost 640,000 in standing time and resequencing — less than the 900,000 assessed impact. R-05 materialised in part, drawing 310,000.

$$\begin{aligned} \text{Contingency drawn} &= 640,000 + 310,000 = 950,000 \\ \text{Remaining contingency} &= 2,600,000 - 950,000 = 1,650,000 \end{aligned}$$

Both draws were posted as transfers into the affected control accounts. BAC remains 26,600,000; the contingency pot inside it fell by 950,000 and control account budgets rose by the same amount.

The comfortable test. Earned value shows the project 40.0 % complete. Contingency drawn is $950,000 \div 2,600,000 = 36.5 \%$. Drawing 36.5 % of the reserve at 40.0 % complete looks disciplined, and this is the figure that appears on most reserve reports.

The test that matters. The register is re-assessed at the same date:

Risk	Status at Month 9	Probability	Impact	EMV
R-01	Occurred and closed; drawn	—	—	—
R-02	Open; trial pits raised the probability	0.45	1,500,000	$0.45 \times 1,500,000 = 675,000$
R-03	Closed — remaining volume price-fixed; contingency released	—	—	0
R-04	Open, unchanged	0.15	2,000,000	$0.15 \times 2,000,000 = 300,000$
R-05	Occurred in part; residual closed	—	—	—
R-06	New: commissioning resource availability	0.25	1,200,000	$0.25 \times 1,200,000 = 300,000$
Remaining EMV				1,275,000

Re-running the model on the remaining register at the same P80 confidence gives a **remaining exposure of 1,900,000**.

$$\begin{aligned} \text{Remaining contingency} &= 1,650,000 \\ \text{Remaining exposure (P80)} &= 1,900,000 \\ \text{Shortfall} &= 1,900,000 - 1,650,000 = (250,000) \end{aligned}$$

The project is 250,000 short of the confidence level it was funded to, at Month 9 of 20 — while the burn ratio says it is fine. The gap opened because R-02's probability rose after the trial pits and a

new risk (R-06) entered the register; both are ordinary events, and neither is visible in the drawdown percentage.

What the report should say. "Contingency remaining 1.65 million against remaining assessed exposure of 1.90 million at P80 — a shortfall of 0.25 million. Driven by R-02, whose probability rose to 0.45 following the trial pit results, and by new risk R-06. Options: fund the gap from management reserve as a baseline change, reduce exposure by advancing the ground investigation to convert R-02 into a known quantity, or accept a reduced confidence level and record the decision. Recommendation: advance the ground investigation at an estimated 140,000, which retires the largest single exposure. Decision required this month; the option closes when the foundation sequence starts."

If management reserve were released. Suppose the sponsor releases 800,000 of management reserve to fund the gap and a small scope addition:

```
New BAC = 26,600,000 + 800,000 = 27,400,000
Management reserve remaining = 1,400,000 - 800,000 = 600,000
Total authorised budget = 27,400,000 + 600,000 = 28,000,000 (unchanged)
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The total authorised budget does not move — the money was always authorised. What moves is the baseline, and therefore every earned value comparison from that date forward. The change log records the release, its date and its reason, and the performance trend is annotated so no reader mistakes the step change for a performance improvement.

Assumptions this example depends on. Probabilities and impacts are the assessed values recorded in the register at each date; the Monte Carlo outputs are model results at the stated confidence level; risk impacts are cost only, with any time-related consequence priced once and not duplicated in schedule contingency; no escalation allowance covers the same exposure as R-03.

10. Checklist

Setting it

- [] Contingency is derived from the risk register, not from a percentage.
- [] The derivation method and confidence level are stated (EMV, or a named P-level).
- [] The gap between the mean and the funded level is disclosed as the cost of confidence.
- [] Shared drivers between risks have been considered, not assumed away.
- [] Escalation, currency and risk allowances do not cover the same exposure twice.

Holding it

- [] Contingency is held at project level, not distributed into control account budgets.
- [] Management reserve is held outside the baseline, by the sponsor.
- [] The budget structure reconciles: control accounts + contingency = BAC; BAC + management reserve = total authorised budget.

Drawing it

- [] Every draw cites a risk identifier and the evidence that the risk occurred.
- [] Approval thresholds by value are defined and followed.
- [] Draws are posted as transfers within the baseline; BAC is unchanged.

- [] Every draw is in the change log with a number, date and approver.
- [] Contingency for risks that have passed is released deliberately, not left in the pot.

Reading it

- [] Remaining contingency is compared with remaining exposure at the same confidence level, monthly.
- [] The drawdown curve is plotted against risk retirement, not against percent complete.
- [] Any shortfall is reported with options and a recommendation, not as a status.
- [] Management reserve releases are annotated on every performance trend they affect.
- [] At completion, undrawn contingency is explained rather than presented as a saving.

— Related

- [BPG-16 – Risk registers that work](#) — the register that contingency must be traceable to; without it, nothing in this guide is possible.
- [BPG-17 – Quantitative schedule risk analysis](#) — where a defensible P-level comes from, and how correlation moves the tail.
- [BPG-04 – Baselining and baseline change control](#) — the procedure for a management reserve release and for logging drawdown transfers.
- [BPG-09 – Estimate at completion: choosing and defending a method](#) — what belongs in a forecast and what sits beside it as held reserve.
- [TPL-10 – Risk register](#) — the instrument, including the contingency allocation columns used in §9.

— Sources and standards

The two-reserve structure and the derivation of contingency from analysed risk are described in published cost and risk management frameworks, among them the AACE International Total Cost Management framework, the PMBOK Guide and the ISO 31000 risk management principles; they are explained here in our own words and no text or table is reproduced. Accounting treatment of provisions and reserves is governed by the applicable financial reporting framework and is outside the scope of this guide. The internal references are BoK Domain 3 (Budgeting & Forecasting) and BoK Domain 12 (Risk Management for Project Controls). All figures in §9 are illustrative and were computed for this document.

— Status and version

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